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United States Patent	12398025
Kind Code	B2
Date of Patent	August 26, 2025
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Winch equipped with a variable torque limiter

Abstract

A winch includes a rotary drum, a cable which can be wound on the drum in multiple layers, and a brake for braking the rotation of the drum, the brake being calibrated to permit the rotation of the drum beyond a variable torque which decreases as the length of the cable wound on the drum increases.

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Appl. No.:	18/266244
Filed (or PCT Filed):	December 17, 2021
PCT No.:	PCT/EP2021/086601
PCT Pub. No.:	WO2022/129577
PCT Pub. Date:	June 23, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240025712 A1	Jan. 25, 2024

Foreign Application Priority Data

FR	2013449	Dec. 17, 2020
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Publication Classification

Int. Cl.: B66D1/58 (20060101); B66D5/12 (20060101); F16D43/21 (20060101)

U.S. Cl.:

CPC B66D1/58 (20130101); B66D5/12 (20130101); F16D43/216 (20130101)

Field of Classification Search

CPC: B66D (5/14); B66D (5/22); B66D (1/58); F16D (43/216)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2017352	12/1934	Quick	254/350	B66D 1/14
3572482	12/1970	Kalpas	192/223.1	B66D 5/00
4175727	12/1978	Clarke	254/274	B66D 5/26
2002/0084154	12/2001	Peter	188/158	B60T 13/746
2008/0190714	12/2007	Pan	188/71.2	B66D 1/58
2014/0083806	12/2013	Miyazaki	188/151R	B66D 5/28
2016/0340158	12/2015	Maghsoodi	N/A	F16D 13/75
2020/0307971	12/2019	Maghsoodi	N/A	B66D 5/14

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
100 34 336	12/2002	DE	N/A
3 028 983	12/2015	EP	N/A
WO-2009062232	12/2008	WO	B66D 1/16

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage of International patent application PCT/EP2021/086601, filed on Dec. 17, 2021, which claims priority to foreign French patent application No. FR 2013449, filed on Dec. 17, 2020, the disclosures of which are incorporated by reference in their entirety.

FIELD OF THE INVENTION

(2) A winch generally comprises a drum around which a cable supporting a load is wound. An excessive force applied to the cable can damage the winch and notably the cable, the drum or its drive device. The invention relates to the safety linked to the extraordinary forces that can act on the cable of a winch.

BACKGROUND

(3) In a winch used to lift the load, the forces transmitted by the cable are generally well controlled. This involves the weight of the load. Once it has been lifted, the force remains substantially

constant, considering that the traction speed is low and does not change much. It is easy to provide protection from disruptive forces by measuring the force, for example using a force sensor. It is then possible to compare the measurement to a reference and inform the user of the winch when the force measured exceeds the reference, enabling the user to react suitably, notably by stopping the winch.

(4) In towing winches, used notably on board ships or helicopters to launch various objects, such as for example sonar antennae, seismic sensors, etc., the force variations on the cable can be very violent and very sudden. These variations can be caused for example by the navigation conditions of the ship, or by the collision of the towed object with an obstacle.

(5) In view of these severe increases in tension on the cable, it is necessary to protect the cable from breaking, since in many cases the cable subjected to an excessive force stretches and thus accumulates potential energy which can be released instantaneously if the cable breaks. This explains the fact that, when breakage occurs, the cable makes very violent whiplash movements. These movements can then damage the blades of a helicopter or the integrity of its cabin, this representing a considerable danger for the helicopter and its occupants. In the case of a ship, the operators who are close to the winch and perform the launch and recovery of the towed objects can also be in danger if the cable breaks.

(6) The protection of persons and equipment must of course be ensured whether the winch is under tension or not.

(7) One way of providing protection against these violent forces is to oversize the winch, its cable and the equipment which surrounds it, in order for them to resist the extraordinary forces to which the cable might be subjected. Oversizing is rarely desirable since it requires a significant increase in mass of the equipment, this being impossible in practice in the case of equipment which is on board a helicopter. The result would be a mass which is incompatible with this type of carrier.

Furthermore, even if it is possible to protect the winch itself, it is virtually impossible to protect the surroundings of the winch and notably the blades of the rotors and the cabin itself.

(8) Another way of providing protection against these excess tensions is to let the cable be paid out by way of a torque limiter on the axle of the winch. The torque limiter permits the drum of the winch to slide in relation to its drive shaft in order to make it possible for the cable to unwind, thus limiting the force to which it is subjected.

(9) However, the limitation of the force on the cable by way of a torque limiter has a drawback. When the cable is wound in multiple layers, the torque from which the limiter operates corresponds to a variable force on the cable depending on the number of layers of the cable on the drum. More specifically, the torque is equal to the force on the cable multiplied by the radius of the winding layer. For a given torque from which the limiter operates, the more the cable is wound on the drum, the more the force exerted on the cable to trigger the torque limiter decreases. Depending on the tolerances of the torque limiter, and on the variation in the winding diameter of the cable on the drum between the first layer in contact with the barrel of the drum and the winding layer, the dimensioning may be impossible. For example, it is possible that the torque for which it is to be ensured that the cable pays out when it is at the barrel of the drum is less than the torque for which it is to be ensured that the cable does not pay out when it is in the last layer on the drum.

SUMMARY OF THE INVENTION

(10) The invention aims to propose a solution for ensuring sliding of the cable for a force which does not vary much, even when the cable can be wound on the drum with a great number of layers.

(11) To that end, the invention relates to a winch comprising a frame, a drum which can rotate in relation to the frame about an axis of rotation, a cable which can be wound on the drum in multiple layers, a brake for braking the rotation of the drum, the brake being calibrated to permit the rotation of the drum beyond a variable torque which decreases as the length of the cable wound on the drum increases, and a mechanism for adjusting the torque exerted by the brake comprising a screw-nut system of which the nut is driven in rotation with the drum, and of which a translational movement

- of the screw driven by the nut is configured to adjust the torque exerted by the brake
- (12) In a particular embodiment, the brake comprises at least one first plate which is movable in translation in relation to the frame along the axis of rotation of the drum and is prevented from rotating in relation to the frame about the axis of rotation of the drum, at least one second plate which is prevented from rotating in relation to the drum about the axis of rotation of the drum, a compression element designed to compress the first plate against the second plate along the axis of rotation of the drum, the adjusting mechanism being configured to adjust the compression of the compression element depending on the rotation of the drum.
- (13) The compression element may be a first elastic element which exerts a force proportional to its length along the axis of rotation of the drum.
- (14) The winch may comprise a mechanism for manoeuvring the brake between two positions, a first position enabling the free rotation of the drum and a second position in which the brake opposes the rotation of the drum up to the variable torque.
- (15) The winch may comprise a motor for making the drum rotate, and a control module permitting the rotation of the motor in the first position of the brake manoeuvring mechanism and preventing the rotation of the motor in the second position of the brake manoeuvring mechanism.
- (16) The brake manoeuvring mechanism is advantageously configured to deactivate the adjusting mechanism in the first position of the brake.
- (17) The brake manoeuvring mechanism may comprise a third plate disposed between the first elastic element and the first plate, the third plate being movable in translation along the axis of the drum, and wherein the brake manoeuvring mechanism is configured to compress the first elastic element by moving it away from the first plate in its first position and to leave the third plate free in its second position.
- (18) The brake manoeuvring mechanism may comprise a second elastic element disposed between the first plate and the third plate.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention will be better understood and further advantages will emerge on reading the detailed description of an embodiment which is given by way of example, this description being illustrated by the appended drawing in which:

(2) FIG. 1 shows a preferred embodiment of a winch according to the invention.

(3) For the sake of clarity, identical elements bear the same references in the various FIGURES.

DETAILED DESCRIPTION

(4) FIG. 1 schematically shows an example of a winch **10** according to the invention. The winch **10** comprises a drum **12** and a cable **14** which can be wound on the drum **12**. The cable **14** can be wound in multiple layers around the drum **12**. FIG. 1 shows two complete layers **16** and **18** and one partial layer **20**, referred to as winding layer. The more the cable **14** is wound, the more the number of layers increases. The drum **12** is rotatable about an axis **22**. The rotation of the drum **12** makes it possible to wind and unwind the cable **14** depending on the direction of rotation. The rotation can be driven by a motor **24**, which is for example disposed inside the drum **12**.

(5) According to the invention, the winch **10** comprises a brake **26** for braking the rotation of the drum **12**. In the example shown, the brake **26** comprises surfaces which come into contact with one another to provide the braking. A first type of surfaces is prevented from rotating in relation to a frame **28** of the winch **10** and a second type of surfaces is prevented from rotating in relation to the drum **12**. Prevention from rotating is understood to mean: suppression of a degree of rotational freedom about the axis **22**. A complete connection makes it possible to prevent the rotation. As an alternative to the complete connection, aside from the suppressed degree of rotational freedom,

other degrees of freedom can remain, notably to enable the positioning of the two types of surfaces in relation to one another by avoiding an excessively tightly fitting chain of dimensions. The two types of surfaces can rub against one another until the drum **12** stops. The invention is interested in parking braking and in the possible sliding of the brake **26** to avoid the cable **14** being subject to excessive tension. The brake **26** may also be used to decelerate the rotation of the drum **12** by converting the kinetic energy of the drum **12** to heat energy by way of the friction between the two surfaces. As an alternative, other types of brakes can be implemented within the context of the invention, for example by converting the kinetic energy of the drum **12** to another energy, for example electrical or pneumatic energy.

(6) The brake **26** is calibrated to permit the rotation of the drum **12** beyond a variable torque which increases as the length of the cable **14** wound on the drum **12** increases. When the drum **12** is at a standstill, for a given number of layers of the cable **14** that is wound on the drum **12**, the traction force on the cable **14** is equal to the resistive torque that keeps the drum **12** at a standstill divided by the radius of the winding layer, the layer **20** in FIG. **1**. If deterioration or even breakage of the cable **14** is to be avoided, it is important to limit the traction force on the cable. To do this, a maximum force above which the cable must be able to “pay out” is defined. In other words, above the maximum force, the drum **12** rotates under the traction of the cable **14** to allow the cable **14** to unwind and thus avoid exceeding the maximum permitted traction force. Owing to the variable number of turns wound on the drum **12**, by varying the maximum torque absorbed by the brake depending on the length of the unwound cable, it is possible to keep the force linked to this torque as constant as possible over the entire length of the cable.

(7) To be as precise as possible and maintain a properly constant force, it would be necessary to vary the maximum torque of the brake **26** depending on the number of the last layer of the wound cable. If the winch **10** has a spooling gear for best arranging the layers on the drum **12**, this spooling gear can transmit the information on the change of layer which makes it possible to increment the maximum torque of the brake. As an alternative, to simplify the winch **10**, it is possible to continuously vary the maximum torque of the brake by approximating the increments. The embodiment of FIG. **1** can be inscribed in this continuous variation of the maximum torque of the brake above which the brake slides by enabling the rotation of the drum **12** and the unwinding of the cable **14**.

(8) In the embodiment of FIG. **1**, the drum **12** is secured to a shaft **30** extending along the axis **22**. The shaft **30** is connected to the frame **28** by way of a rotary bearing **32**, shown schematically in FIG. **1**. This may involve rolling bearings that carry the assembly of the drum **12** and its shaft **30**. Any other type of bearing is possible. The shaft **30** may also be used to transmit energy from the motor **24** to the drum **12**.

(9) In the example shown, the brake **26** comprises three plates **34**, **36** and **38** which are movable in translation in relation to the frame **28** along the axis **22**. The plates **34**, **36** and **38** are prevented from rotating in relation to the frame **28** about the axis **22**. The brake also comprises two plates **40** and **42** which are prevented from rotating in relation to the shaft **30**. At least one of the plates **40** or **42** is free to move in translation in relation to the shaft **30**. The plates **34**, **36** and **38**, which are prevented from rotating in relation to the frame **28**, and the plates **40** and **42**, which are prevented from rotating in relation to the shaft **30**, alternate. To ensure the braking of the drum **12**, the plates **34**, **36** and **38** sandwich the plates **40** and **42**. In practice, the invention can be implemented irrespective of the number of plates that are prevented from rotating in relation to the frame **28** and the number of plates that are prevented from rotating in relation to the shaft **30** provided that these plates face one another and that a means for pressurizing these two types of plates is provided. The resistive torque that the brake **26** can bring about to oppose the rotation of the drum **12** is proportional to the surface area and the number of interfaces between the two types of plates and to the force exerted to press the plates against one another. As a result, to limit the force that is to be exerted, it can be advantageous to increase the number of plates of the two types. In the example

shown, there is one more plate that is prevented from rotating in relation to the frame **28** than there are plates that are prevented from rotating in relation to the shaft **30**. In the example shown, two plates that are prevented from rotating in relation to the shaft **30**, the plates **40** and **42**, are used. In order to enable the compression of the plate **36**, the plates **40** and **42** should have a degree of freedom in translational movement in relation to the shaft **30** whilst still being prevented from rotating in relation to the shaft **30**. The connection between the plates **40**, **42** and the shaft **30** can be provided by grooves extending along the axis **22**. Furthermore, the plates **40** and **42** are prevented from rotating in relation to the shaft **30**, which is itself secured to the drum **12**. It would also be possible for one of the flanges of the drum **12** to perform the function of the two plates **40** and **42**.

(10) The various plates **34** to **42** have for example a circular cross section around the axis **22**. The two types of plates **34**, **36** and **38**, for the one part, and **40** and **42**, for the other part, have the same diameter, this simplifying the production of the brake **26**. Any other shape of the plate is possible. It is even possible to provide multiple plates of one of the types facing a single plate of the other type. It is, for example, possible to provide a pair of shoes, which are prevented from rotating in relation to the frame **28** about the axis **22**, which compress a disk secured to the drum **12**. Multiple pairs of shoes can be implemented around one and the same disk.

(11) The brake **26** comprises a compression element designed to compress the plates **34** to **42** against one another. In FIG. **1**, the compression element **44** is shown in the form of multiple helical springs. A single helical spring wound around the axis **22** can also be implemented. The compressive force is proportional to the length L of the elastic element **44** along the axis **22**. Any other compression element may also be used. The compression can for example be hydraulic or pneumatic, with the compressive force being proportional to the pressure of the fluid used.

(12) The brake **26** also comprises a mechanism **46** for adjusting the torque exerted by the brake. In the example shown in FIG. **1**, the adjusting mechanism **46** makes it possible to adjust the force exerted by the compression element **44** depending on the rotation of the drum **12**. More specifically, when the drum **12** rotates so as to unwind the cable **14**, the number of layers of the cable **14** wound on the drum **12** decreases and the adjusting mechanism **46** is configured such that the force exerted by the compression element **44** decreases so as to reduce the maximum torque able to be exerted by the brake **26** before it slides. The variation of the force exerted by the compression element **44** depending on the rotation of the drum **12** is defined so as to keep the force exerted on the cable **14** in accordance with the maximum torque of the brake **26** substantially constant or at the very least substantially constant between two successive winding layers of the cable **14** all along its winding around the drum **12**.

(13) When the compression element **44** is an elastic element of which the length L is to be varied depending on the rotation of the drum **12**, the adjusting mechanism **46** is configured to convert a rotational movement of the drum **12** into a translational movement of one of the ends of the elastic element **44**. A screw-nut system is well suited to this conversion of movement. Other movement conversion mechanisms are also possible, such as a cam mechanism or an eccentric mechanism. It is also possible to implement a double-acting linear cylinder for displacing the adjustable end of the elastic element **44**. The chambers of the cylinder are supplied by a two-way pump actuated by the rotation of the shaft **30**. To return to the screw-nut system, the shaft **30** may be threaded and a nut interacting with this thread may bear against the elastic element **44** to adjust the length L of the latter. The thread makes it possible to establish a linear function between the angular position of the drum **12** about its axis **22** and the maximum torque of the brake. The pitch of the thread defines the gradient of the linear function.

(14) It is possible that the desired gradient requires a thread pitch which is too small to be easily produced on the shaft **30** and on the associated nut. It is then possible to provide a reduction in speed between the rotation of the drum **12** and that of the nut, as shown in FIG. **1**. More

specifically, the adjusting mechanism **46** comprises a pinion **48** secured to the shaft **30** and a toothed wheel that forms a nut **50** and meshes with the pinion **48**. The number of teeth of the pinion **48** is smaller than that of the nut **50**, this enabling a reduction in speed between the drum **12** and the nut **50**. A screw **52** interacts with the nut **50** and bears against a chock **54** against which the elastic element **44** bears. The rotation of the drum **12** displaces the chock **54** in translation along the axis **22**. As an alternative to a simple geartrain like that formed by the toothed wheel **48** and the nut **50**, any other speed reduction mechanism is of course possible between the shaft **30** and the nut **50**. It is possible for example to provide a chain mechanism, a mechanism comprising multiple geartrains, a planetary geartrain, etc.

(15) Other forms of brakes can be implemented within the context of the invention. It is for example possible to implement a brake usually referred to as a drum brake, wherein an inner cylindrical surface of axis **22** rotates with the shaft **30**. On the inside of this surface, shoes which can move radially in relation to the axis **22** bear against the inner cylindrical surface. The shoes are rotatably connected to the frame **28** about the axis **22**. The braking torque is a function of the bearing force of the shoes against the inner cylindrical surface. The torque adjusting mechanism then radially displaces the shoes to modify the torque exerted by the brake. In the torque adjusting mechanism, there is a screw-nut system. The translational movement of the screw is radial and the nut driving the screw rotates about the radial axis of the screw and is moved by the shaft **30** via an angular member.

(16) The brake **26** is useful when the winch **10** is at a standstill, in other words when the drum **12** is not rotating and when a force lower than the maximum acceptable force is exerted on the cable **14**. In this situation, the motor **24** is at a standstill. When the winch **10** is to be manoeuvred by actuating the motor **24**, it is useful to release the brake **26**. To that end, the winch **10** comprises a brake manoeuvring mechanism **60**. The mechanism **60** has two end positions: a first position, with the brake released, makes it possible for the drum **12** to freely rotate and a second position, with the brake applied, in which the brake **26** opposes the rotation of the drum **12**. It is possible to control the mechanism **60** continuously between the two end positions in order to gradually brake or release the drum **12**. It is in the position in which the brake **26** is applied that the plates **34** to **42** are compressed against one another and that the drum **12** remains immovable as long as the given variable torque has not been reached. When the brake **26** is released, the motor **24** can act as torque limiter by permitting the cable **14** to pay out if an excessive force is applied to it. Advantageously, the mechanism **60** is coupled to the limiter of the variable torque exerted by the elastic element **44**. More specifically, the brake manoeuvring mechanism **60** is configured to deactivate the compression adjusting mechanism **46** of the first elastic element **44** in the first position of the brake **26**.

(17) An example of a brake manoeuvring mechanism **60** is shown in FIG. 1. The mechanism **60** comprises a plate **62** disposed between the elastic element **44** and the plate **34**. The plate **62** is movable in translation along the axis **22**. In a movement toward the left in FIG. 1, the plate **62** compresses the spring **44** by moving it away from the plate **34**. In other words, the plate **62** prevents the spring **44** from compressing the plates **34** to **42** together. The movement of the plate **62** toward the left releases the brake **26**. Conversely, to reapply the brake **26**, the plate **62** is left free and the spring **44** can compress the plates **34** to **42** together.

(18) The brake manoeuvring mechanism **60** may additionally comprise another elastic element **64** disposed between the plate **34** and the plate **62**. The elastic element **64** also deforms along the axis **22**. When the brake **26** is applied, the plate **62** is allowed to move freely in translation along the axis **22** and the elastic element **64** fully transmits the force exerted by the elastic element **44**. By contrast, when the plate **62** is displaced toward the left, the elastic element **64** expands and no longer transmits the force exerted by the elastic element **44** in order to release the brake **26**. In this position of the brake, the elastic element **64** is preloaded so as to exert only a minimum force on the plate **34**. The elastic element **64** makes it possible simply to compensate for the functional

clearances between the plates **34** to **42** with as little compression as possible.

(19) As an alternative to the disposition of the plate **62** and the elastic element **64** shown in FIG. **1**, it is possible to position a brake manoeuvring device on the right-hand side of the plates **34** to **38**, enabling a translational movement of the plate **38** between a position with the brake applied in which the elastic element **44** bears against the plate **34**, and a position with the brake released in which the elastic element **44** bears to virtually no extent, allowing for the compensation of the functional clearances. However, this disposition of the brake manoeuvring device on the right-hand side makes it more difficult to produce the brake manoeuvring device.

(20) The winch **10** may comprise a control module **70** which controls both the motor **24** and the brake manoeuvring mechanism **60** in a coordinated manner. More specifically, the control module **70** permits the rotation of the motor **24** when the brake is released and prevents the rotation of the motor **24** when the brake is applied.

Claims

1. A winch comprising a frame, a drum which can rotate in relation to the frame about an axis of rotation, a cable which can be wound on the drum in multiple layers, a brake for braking the rotation of the drum, the brake being calibrated to permit the rotation of the drum beyond a variable torque which decreases as the length of the cable wound on the drum increases, and a mechanism for adjusting the torque exerted by the brake comprising a screw-nut system of which the nut is driven in rotation with the drum, and of which a translational movement of the screw driven by the nut is configured to adjust the torque exerted by the brake.
2. The winch as claimed in claim 1, wherein the brake comprises at least one first plate which is movable in translation in relation to the frame along the axis of rotation of the drum and is prevented from rotating in relation to the frame about the axis of rotation of the drum, at least one second plate which is prevented from rotating in relation to the drum about the axis of rotation of the drum, a compression element designed to compress the first plate against the second plate along the axis of rotation of the drum, and wherein the adjusting mechanism is configured to adjust the compression of the compression element depending on the rotation of the drum.
3. The winch as claimed in claim 2, wherein the compression element is a first elastic element which exerts a force proportional to its length (L) along the axis of rotation of the drum.
4. The winch as claimed in claim 2, comprising a mechanism for manoeuvring the brake between two positions, a first position enabling the free rotation of the drum and a second position in which the brake opposes the rotation of the drum up to the variable torque, wherein the brake manoeuvring mechanism is configured to deactivate the adjusting mechanism in the first position of the brake.
5. The winch as claimed in claim 4, wherein the brake manoeuvring mechanism comprises a third plate disposed between the first elastic element and the first plate, the third plate being movable in translation along the axis of the drum, and wherein the brake manoeuvring mechanism is configured to compress the first elastic element by moving it away from the first plate in its first position and to leave the third plate free in its second position.
6. The winch as claimed in claim 5, wherein the brake manoeuvring mechanism comprises a second elastic element disposed between the first plate and the third plate.
7. The winch as claimed in claim 1, comprising a mechanism for manoeuvring the brake between two positions, a first position enabling the free rotation of the drum and a second position in which the brake opposes the rotation of the drum up to the variable torque.
8. The winch as claimed in claim 7, comprising a motor for making the drum rotate, and a control module permitting the rotation of the motor in the first position of the brake manoeuvring

mechanism and preventing the rotation of the motor in the second position of the brake manoeuvring mechanism.
